

Module 11: Queues, Jobs & Scheduling

Duration: 4-5 days | **Level:** Intermediate-Advanced

Why Queues?

Offload slow work (emails, API calls, image processing) to background workers so HTTP responses stay fast.

Creating Jobs

```
php artisan make:job SendTaskDueReminder

class SendTaskDueReminder implements ShouldQueue
{
    use Queueable;

    public function __construct(public Task $task) {}

    public function handle(): void
    {
        Notification::send($recipients, new TaskDueReminder($this->task));
    }
}
```

Dispatch:

```
SendTaskDueReminder::dispatch($task);
SendTaskDueReminder::dispatch($task)->delay(now()->addMinutes(5));
```

Queue Configuration

```
QUEUE_CONNECTION=database    # Good for learning
# QUEUE_CONNECTION=redis      # Production recommended

php artisan queue:table
php artisan migrate
php artisan queue:work
php artisan queue:work --tries=3 --timeout=60
```

Task Scheduler

routes/console.php:

```
Schedule::command('tasks:send-reminders')->dailyAt('08:00');
```

Cron entry (production):

```
* * * * * cd /var/www/taskforge && php artisan schedule:run >> /dev/null 2>&1
```

Local testing:

```
php artisan schedule:work
php artisan tasks:send-reminders
```

Failed Jobs

```
php artisan queue:failed  
php artisan queue:retry all
```

TaskForge Files

- app/Jobs/SendTaskDueReminder.php
 - app/Console/Commands/SendDueReminders.php
 - app/Notifications/TaskDueReminder.php (also queued)
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Exercises

1. Create ProcessAttachment job for virus scanning simulation.
 2. Configure QUEUE_CONNECTION=sync vs database and compare behavior.
 3. Add retry logic with \$tries = 3 and \$backoff = [10, 60, 300].
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Next Module

-> [Module 12: Events & Listeners](#)